

12). 由(1) 知此时 $R_1 = R_w$

$$\therefore U_0 = 100 \text{ mV}$$

$$(3) \therefore U_0 = 10 \frac{R_w}{R_1} (U_{I2} - U_{I1}) = \frac{100 \text{ k}\Omega}{10 \text{ k}\Omega - R_2} (U_{I2} - U_{I1})$$

$$|U_0| = \frac{100}{10 - R_2} |U_{I2} - U_{I1}| \leq 14, |U_{I2} - U_{I1}|_{\max} = 20 \text{ mV}$$

$$\therefore R_2 \leq 9.86 \text{ k}\Omega, (R_2)_{\max} = 9.86 \text{ k}\Omega$$

$$6.10. (a) U_{I1} \text{ 接地}, U_{O2} = - \frac{R_3 + R_4}{R_2} (1 + \frac{R_3 // R_4}{R_5}) U_{I2};$$

$$U_{I2} \text{ 接地}, U_{O1} = - \frac{R_3 + R_4}{R_1} (1 + \frac{R_3 // R_4}{R_5}) U_{I1}$$

$$\therefore U_0 = - (R_3 + R_4) (1 + \frac{R_3 // R_4}{R_5}) (\frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2})$$

(b) $U_{O1} = (1 + \frac{R_3}{R_1}) U_I$, 假设加在 A_1 同相输入端的信号为 U_{I1} , 加在 A_2 同相输入端的信号为 U_{I2} ,

$$U_I = U_{I1} - U_{I2}$$

$$\text{则 } U_{O1} = (1 + \frac{R_3}{R_1}) U_{I1}$$

$$U_0 = \frac{R_5}{R_3} U_{I2} - \frac{R_5}{R_4} U_{O1}$$

$$= (1 + \frac{R_5}{R_4}) U_{I2} - \frac{R_5}{R_4} (1 + \frac{R_3}{R_1}) U_{I1}$$

$$= \frac{R_5}{R_4} (1 + \frac{R_5}{R_4}) (U_{I2} - U_{I1})$$

$$= - (1 + \frac{R_5}{R_4}) U_I$$

(c) A_1, A_2, A_3 的输出信号分别为 U_{I1}, U_{I2}, U_{I3}

$\therefore$ 对 A_4 ,

$$\begin{cases} \frac{U_{I1} - U_+}{R_1} + \frac{U_{I2} - U_+}{R_{a1}} + \frac{U_{I3} - U_+}{R_1} = \frac{U_+}{R_2} \\ \frac{U_0 - U_-}{R_4} = \frac{U_-}{R_3} \\ U_+ = U_- \end{cases}$$